

Trainings ▾

General Information

Jacket

CAUTION

The seal will malfunction if the spring tension is not correct. One drop of Loctite® 290, or equivalent, must be applied on the joint between the seat and the shaft. More than one drop of the sealant can contact the sealing surfaces and cause the seal to malfunction.

CAUTION

Use of the K50 water pump inlet connection on low temperature aftercooler (LTA) equipped engines will result in excessive cooling system pressures and cooling system malfunctions.

The K38 and K50 water pumps are the same physical size but contain different parts. The correct water pump **must** be installed. If the correct parts are **not** installed, the performance of the water pump will **not** be within specifications.

The water pump is a centrifugal pump with cast iron impellers **only**. The water pump is gear-driven by a spline coupling. This coupling connects the pump to the water pump drive.

The water pump contains two ball-type bearings. The bearings are lubricated with pressurized engine oil. The new K38 and K50 water pumps contain a front large ball bearing that is a single row type bearing. Older pumps contain a double row type bearing. The water pump body and the bearing spacer are different for the single and double row bearings. If the front bearing is replaced, the same type of bearing **must** be installed.

The water pump shaft and impeller were changed to increase the torque capability of the shaft and impeller joint. The bore in the impeller is smaller. The drive key, the notch for the drive key, and the self-locking capscrew in the impeller were omitted. The older and the new shafts and impellers can be combined, but the torque capability of the joint will **not** be increased. If older and new parts are combined, the drive key and the self-locking capscrew **must** be omitted.

The impeller and the water inlet connection have a machined angle that is critical to the performance of the pump. The older and the new K38 parts do **not** have the same angle and are **not** to be combined. The K50 impeller and water inlet connection are always machined to the same angle, but are different than the older and new K38 parts.

The water pump shaft was also changed. The new shaft contains one retaining ring **only** and requires a spacer between the two bearings. The spacer is different for the single row and double row, front, large bearing. The older shaft contains two retaining rings and does **not** require a

spacer. The older shaft can be installed **only** when the double row bearing is required. If the older shaft is replaced, the correct bearing and spacer **must** be installed for the water pump body.

The water pump contains an oil seal and a water pump seal. A revised water seal was introduced into production in October, 1997 (Engine Serial Number First 33139138). The cavity between these seals has a vent. The vent prevents contamination of the lubricant or coolant if a seal leaks. Oil and coolant seeping from the vent will **not** harm the operation of the pump. Check the vent for any obstruction at each scheduled maintenance interval. The recommended interval is 250 hours or 6 months of operation.

The water seal was changed from a two piece to a unitized, one piece design. The seal and seat are fastened together in this design. This helps to prevent malfunctions caused by dirt or oil contacting the seal and seat surfaces during installation. A special tool **must** be used to install the one piece seal so that the seat is installed at the correct dimension from the water pump body. Correct installation results in proper spring tension.

Additional information regarding other parts of the cooling system can be found within this section.

K50 engines fitted with an LTA system use the K38 water pump and water pump inlet connection.

The water pump inlet connection is designed for optimum coolant flow split and decreases the possibility of reverse flow through the main radiator core via the bypass tube. A revised larger diameter water bypass tube is also used.

The following table details the compatibility of the water pump parts.

Note : If the shaft does not contain threads in the end for the impeller, the drive key **must** be omitted.

Note : If the impeller does not contain a slot on the side that has vanes, the drive key and self-locking capscrew **must** be omitted.

Note : If the shaft contains threads in the end for the impeller, and the impeller contains a slot on the side that has vanes, the drive key and self-locking capscrew **must** be installed.

Preparatory Steps

Jacket

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the battery cables. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Remove the coolant bypass tube. Refer to Procedure 008-005 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-005-tr.html)

Dual Impeller

⚠ WARNING ⚠

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⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

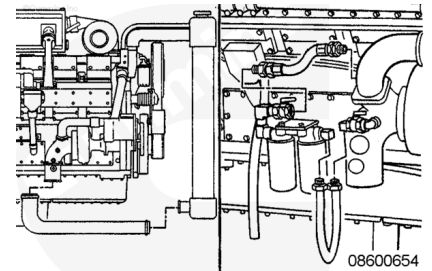
- Disconnect the battery cables. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13.
(/qs3/pubsys2/xml/en/procedures/28/28-013-005-tr.html)
- Remove the alternator drive pulley. Refer to Procedure 009-010 in Section 9.
(/qs3/pubsys2/xml/en/procedures/28/28-009-010-tr.html)
- Remove the water pump bypass and feed tubes. Refer to Procedure 008-138 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-138.html)

Remove

Jacket

Disconnect the lower radiator hose from the water pump inlet.

Note : On LTA-equipped engines, disconnect the water filter hoses.

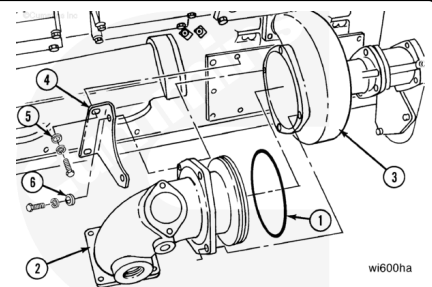


Remove the two capscrews and heavy washers (6) that hold the support bracket (4) to the cylinder block.

Remove the four capscrews that hold the housing (2) to the water pump (3). Note that the two washers (5) are heavy washers.

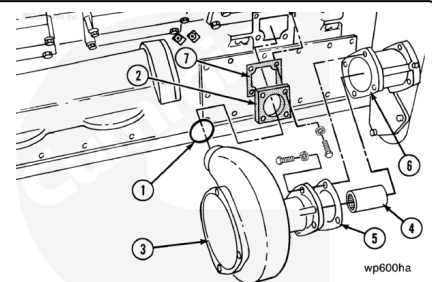
Remove the support bracket. Pull the housing from the pump.

Remove and discard the o-ring (1).



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the four capscrews that hold the adapter plate (2) to the cylinder block.

Remove the four capscrews that hold the pump (3) to the water pump drive (6). Remove the pump and the coupling (4).

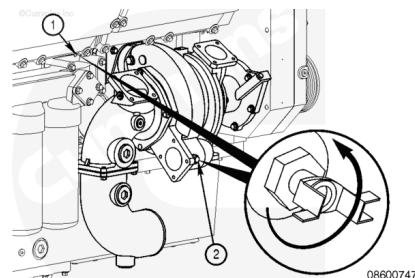
Remove the plate (2) from the pump.

Remove and discard the o-ring seal (1) and the gasket (5).

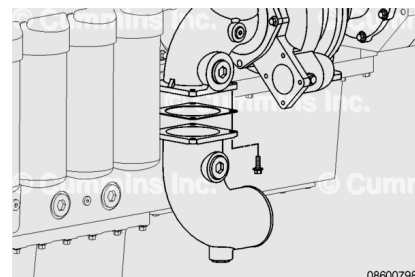
Dual Impeller

Note : The procedure for the QSK60 engine is shown. The procedure for the QSK38 and QSK50 engine is similar.

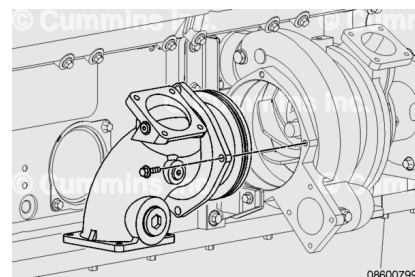
Open the draincock on the engine block (1) and the water pump (2).



Remove the four capscrews and jacket water radiator inlet connection.



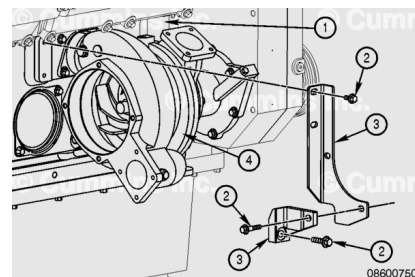
Remove the capscrews and jacket water inlet connection.



Remove the two capscrews (2) from the engine block (1) and the water pump support bracket (3).

Remove the two capscrews (2) from the water pump (4), and the water pump support bracket (3).

Remove the capscrews (2) and the water pump support bracket (3) from the water pump (4), and the engine block (1).



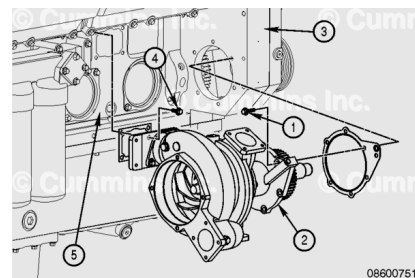
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Remove the six capscrews (1) from the water pump (2) and the gear housing (3).

Remove the four capscrews (4) from the engine block (5).

Remove the water pump from the front gear housing.

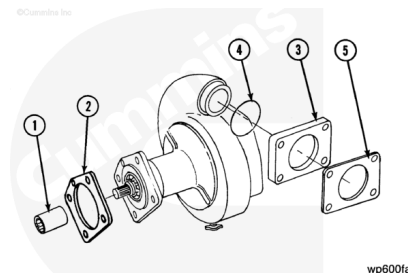


Clean and Inspect for Reuse

Jacket

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturers' recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

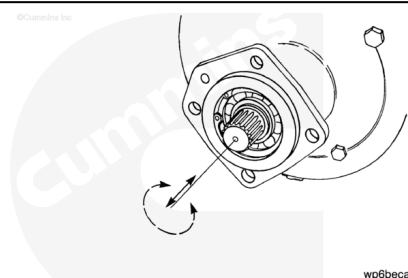
Clean all gasket surfaces.

Use solvent to clean the drive shaft end of the pump.

Dry with compressed air.

Rotate the shaft to check the bearings for damage or wear. An end clearance of 0.05 to 0.010 mm [0.002 to 0.004 in] is required.

If the bearings are damaged or worn, the pump **must** be rebuilt.

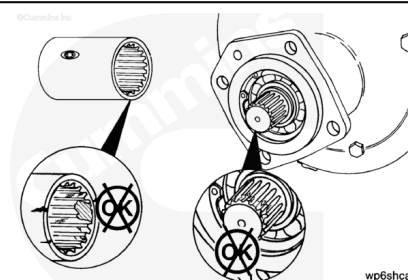


Check the splines on the shaft and the coupling for damage or wear.

If the splines on the shaft are worn or damaged, the pump **must** be rebuilt.

If the splines in the coupling are worn, the coupling **must** be replaced.

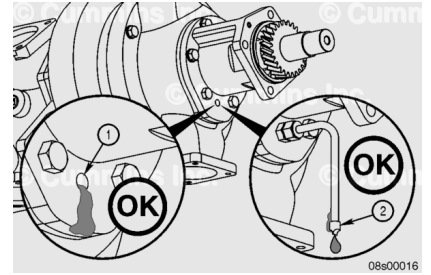
If the splines that contact the water pump drive are worn, check the shaft in the drive.



Dual Impeller

A minor chemical build-up or streaking at the water pump weep hole (1) is normal. Do **not** replace the water pump unless an actual leak is confirmed. If a leak is confirmed, contact a Cummins® Authorized Repair Location for replacement.

Note : Engines with a weep hole filter (2) installed may experience an occasional drip and minor chemical buildup from the filter. This is normal and does not indicate a leaking pump. Do not replace the water pump unless an actual leak is confirmed.

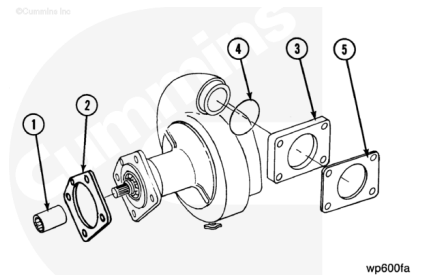


Disassemble

Jacket

Remove the following parts:

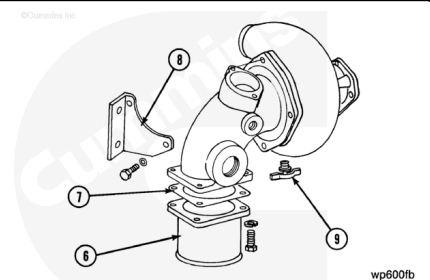
- Spline coupling (1)
- Mounting gasket (2)
- Adapter plate (3)
- O-ring (4)
- Adapter plate gasket (5).



Discard the o-ring and gasket.

Remove the following parts:

- Water inlet connection (6)
- Gasket (7)
- Support bracket (8)
- Draincock (9).

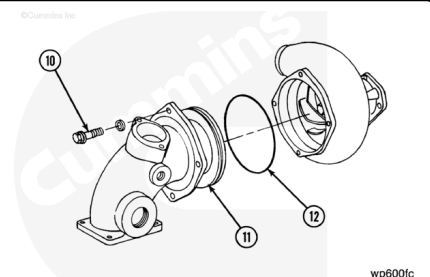


Discard the gasket.

Remove the following parts:

- Capscrews (10)
- Water pump inlet connection (11)
- O-ring seal (12).

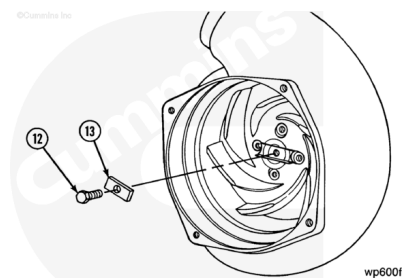
Discard the o-ring.



Note : The latest water pumps and rebuilt water pumps with revised shafts or impellers do not contain the drive key and capscrew.

Remove the following parts:

- Self-locking capscrew (12)
- Drive key (13).

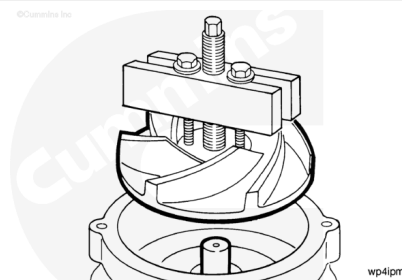


⚠ CAUTION ⚠

The puller screw must pass easily through the impeller bore to prevent damage.

⚠ CAUTION ⚠

The puller screw must not damage the threads in the shaft if the shaft is to be installed again with an impeller that has a slot for a drive key. Install a short capscrew in the shaft to protect the threads.

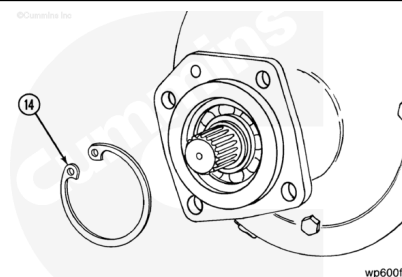


Use a standard puller, Part Number ST-647, or equivalent.

Remove the impeller.

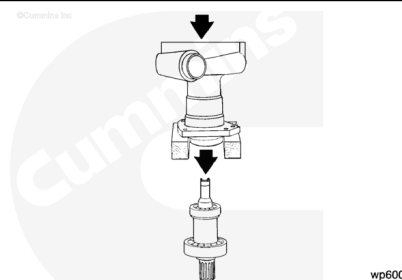
Clean and check the impeller for damage.

Remove the large retaining ring (14).



Use an arbor press to support the water pump body. Push on the impeller end of the shaft. Remove the bearing and shaft assembly.

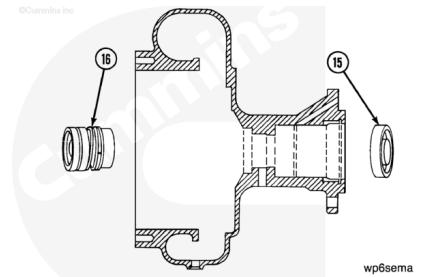
Discard the water pump seal seat, if equipped.



Remove the following parts:

- Oil seal (15)

- Water seal (16).



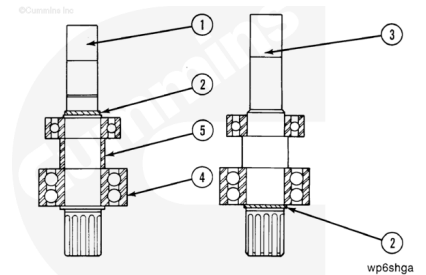
The bearing and shaft assembly can be one of two different designs. The replacement procedure is different for the two designs. The revised bearing and shaft assembly (1), contains a retaining ring (2) nearest to the impeller end of the shaft.

The previous bearing and shaft assembly (3), contains a retaining ring (2) nearest to the end of the shaft that is splined.

The revised assembly (1) can contain different large bearings (4), and bearing spacers (5). The bearing (4) can be a double or single row bearing. The spacer (5) for the single row bearing is longer.

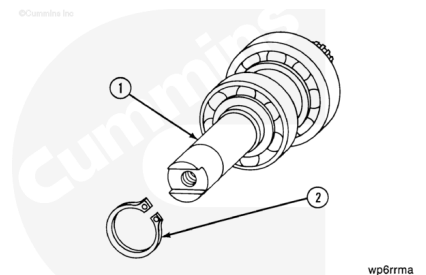
The Water Pump, General Information section in this procedure lists the correct parts.

The previous assembly (3) **must** contain the double row bearing.



The following steps are for the revised design of the water pump.

Remove the retaining ring (2).

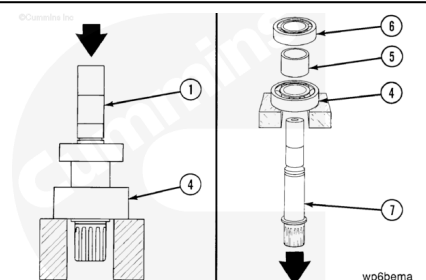


⚠ CAUTION ⚠

Do not damage the end of the shaft if it is to be installed again. Support the assembly as shown.

Use a hydraulic press to support the large bearing (4) as shown.

Remove the small bearing (6), spacer (5), and the large bearing (4) from the shaft (7).

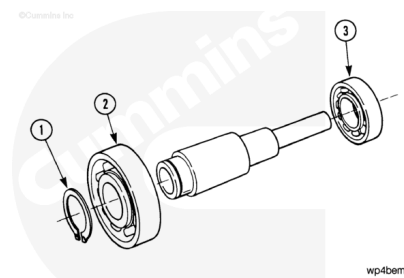


The following step is for the previous design of the water pump.

Remove the retaining ring (1).

Use a water pump bearing separator, Part Number 3375326, or equivalent. Disassemble the parts.

Remove the front large bearing (2), spacer (5), and the rear small bearing (3)



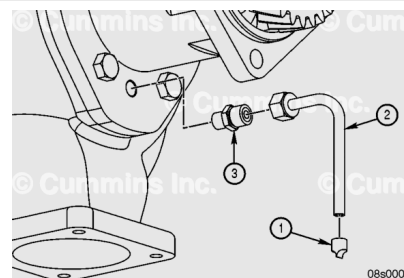
Dual Impeller

Note : There are no replacement parts or disassembly processes for the dual impeller water pump assembly.

Remove the filter (1) from the water pump weep hole drain tube.

Remove the drain tube (2) from the weep hole fitting (3).

Remove the fitting (3) from the water pump.



Inspect

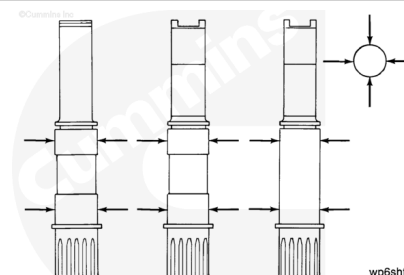
Jacket

Note : Three different shafts have been used. The outside diameter in the bearing area is the same.

Measure the outside diameter

Water Pump Shaft Outside Diameter (Bearing Area)		
mm		in
30.010	MIN	1.1816
30.020	MAX	1.1820

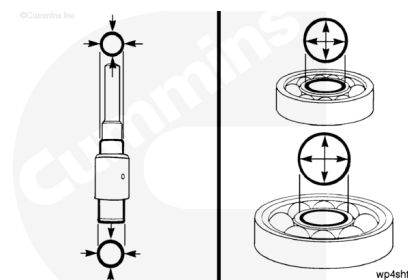
Water Pump Shaft Outside Diameter (Impeller End)		
mm		in
24.643	MIN	0.971
24.653	MAX	0.968



Measure the shaft outside diameter and the bearing inside diameter

Measurements		
	mm	in
Maximum Clearance	0.003	0.0001

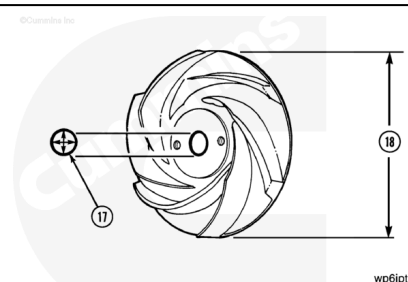
Measurements		
	mm	in
Maximum Interference	0.018	0.007



Note : The shaft outside diameter specifications are the same as for the revised design shaft.

Measure the bore inside diameter.

Water Pump Impeller Inside Diameter (17)		
mm		in
24.563	MIN	0.967
24.589	MAX	0.968

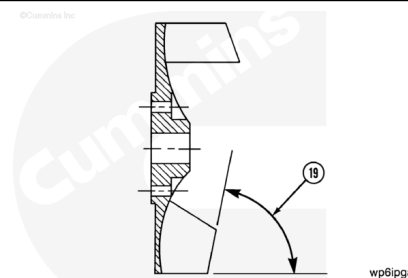


Note : The outside diameter is given for parts identification only. It is not necessary to measure the part.

Water Pump Impeller Outside Diameter (18)			
	mm		in
K38	146	NOM	5.75
K50	171	NOM	6.75

Note : The angle is given for parts identification only. It is not necessary to measure the part.

Note : All K50 water pump impellers have a 78 degree angle.



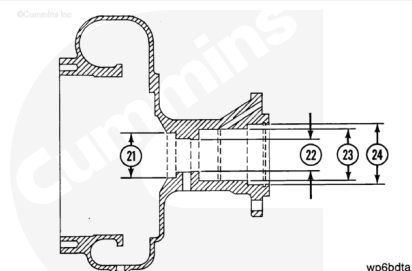
K38 Water Pump Impeller Angle (19)	
Part Number	Angle
3050455	74 degrees
3175069	74 degrees
3050453	67 degrees
3001711	67 degrees

Install an impeller with a 74 degree angle **only** with a water pump inlet connection that has a 16 degree angle.

Install an impeller with a 67 degree angle **only** with a water pump inlet connection that has a 23 degree angle.

K38 and K50 Water Pump Body Bore Inside Diameter

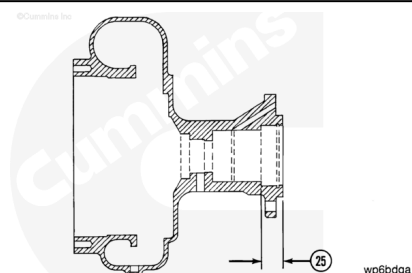
	mm		in
Water Seal (21)	53.60	MIN	2.110
	54.10	MAX	2.130
Oil Seal (22)	38.05	MIN	1.498
	38.10	MAX	1.500
Water Seal (23)	61.997	MIN	2.4408
	62.009	MAX	2.4412
Oil Seal (24)	71.996	MIN	2.8345
	72.012	MAX	2.8351



Note : The depth of the bore is given for parts identification **only**. It is **not** necessary to measure the part.

Front Water Pump Bearing Bore Depth in Water Pump Body (25)

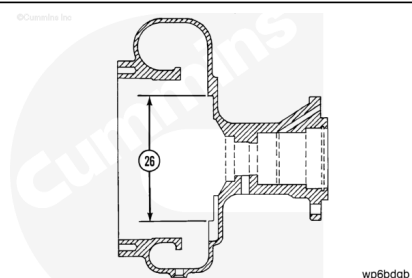
	mm	in
Body for single row bearing	25.4	1.00
Body for double row bearing	36.8	1.45



Note : The diameter is given for parts identification **only**. It is **not** necessary to measure the part.

Water Pump Impeller Counterbore Inside Diameter in Water Pump Body (26)

Engine Model	mm	in
K38 (revised)	147	5.8
K38 (previous)	160	6.3



Water Pump Impeller Counterbore Inside Diameter in Water Pump Body (26)

Engine Model	mm	in
K50 (all)	173	6.8

Assemble

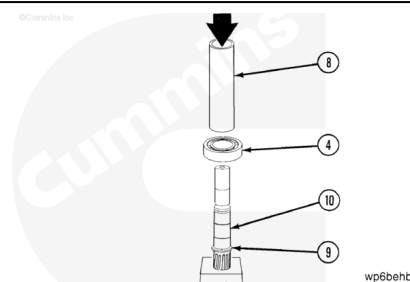
Jacket

⚠ CAUTION ⚠

Be sure the correct large bearing is installed. If the wrong large bearing is installed, the pump will fail.

⚠ CAUTION ⚠

The mandrel must contact on the inner race of the bearing to prevent damage.



Note : On the revised shafts, the bearing will not fit tightly on the part of the shaft that is machined smaller (10).

The following steps are for the revised design shaft and bearings.

Use a hydraulic press to support the shaft as shown.

Use a water pump bearing mandrel, Part Number 3375318 (8), or equivalent.

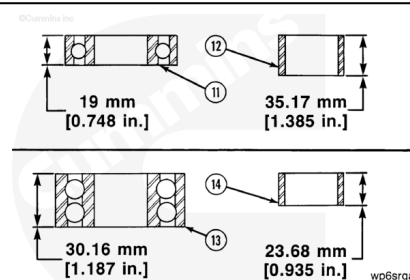
Push the large bearing (4) on the shaft until it touches the step on the shaft (9).

⚠ CAUTION ⚠

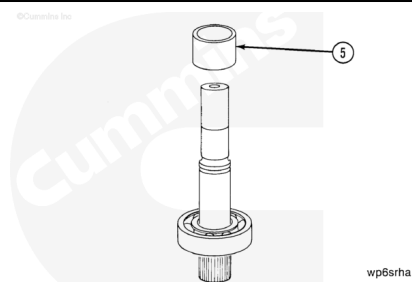
The correct spacer must be installed or the bearing assembly will not fit in the body correctly.

If the large bearing is a single row bearing (11), install the long spacer (12).

If the large bearing is a double row bearing (13), install the short spacer (14).



Install the correct spacer (5). Slide the spacer on the shaft until it touches the large bearing.



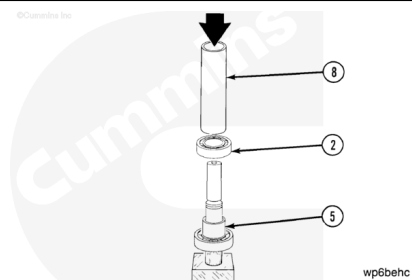
⚠ CAUTION ⚠

The mandrel must contact on the inner race of the bearing to prevent damage.

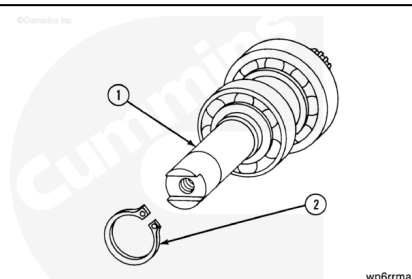
Use a hydraulic press to support the shaft as shown.

Use a water pump bearing mandrel, Part Number 3375318 (8), or equivalent.

Push the small bearing (2) on the shaft until it touches the spacer (5).



Install the retaining ring (2).



⚠ CAUTION ⚠

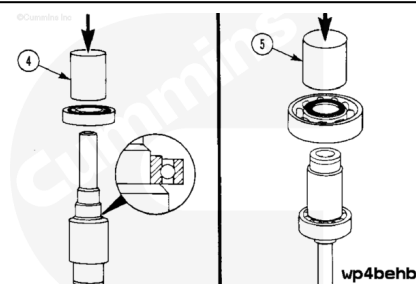
The mandrel must contact on the inner race of the bearing to prevent damage.

Note : Only the large, double row front bearing can be installed on the older design shaft.

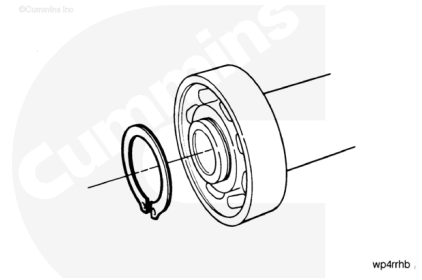
The following steps are for the previous design shaft and bearings.

Use a water pump bearing mandrel, Part Number 3375318 (8), or equivalent. Use an appropriate mandrel (4). Use an arbor press. Support the shaft.

Install the bearings.

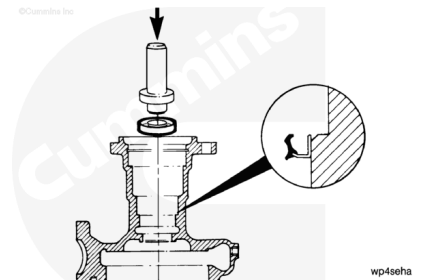


Install the retaining ring.



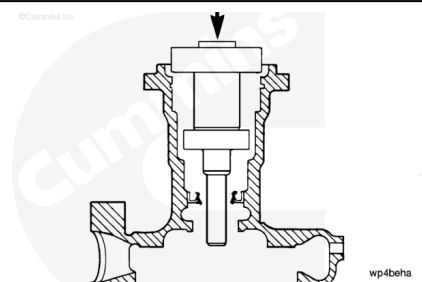
Use water pump seal mandrel, Part Number 3375319, to install the oil seal with the stepped lip facing down toward the impeller.

The seal **must not** be more than 0.51mm [0.020 in] below the top step in the water pump housing.

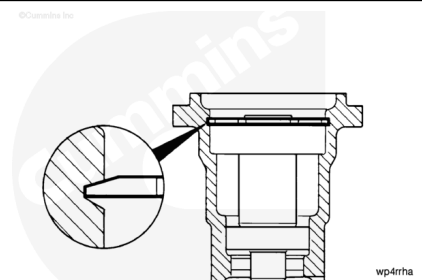


Use an arbor press to support the housing.

Use the water pump bearing driver, Part Number 3375318, or equivalent, or push on the end of the shaft. Install the assembly.



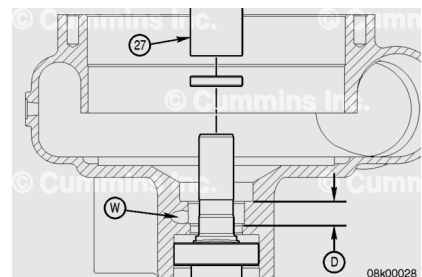
The beveled edge **must** be positioned as shown. Install the large retaining ring.



⚠ CAUTION ⚠

When installing the seal, take care not to damage the seal lip.

Press the oil seal onto the shaft using an oil seal mandrel. If an oil seal mandrel is **not** available, press the seal onto the shaft backwards to open up the seal lip, then turn the seal the correct way around.



Use a water pump seal installer (27) to press the seal into the housing past the weep hole cavity.

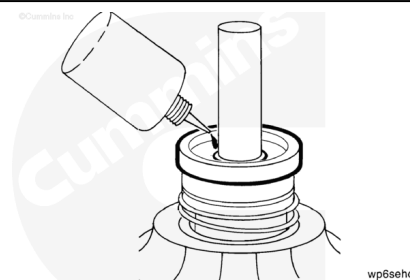
Reference the Tool Catalog, Publication Number 3377710, to determine which tool number to use based on the water pump seal part number.

The installed distance (D) between the oil seal face from impeller side and the inner face in the impeller housing **must** be between 15.60 to 16.79 mm [0.614 to 0.661 in].

Check after installation that the oil seal is **not** obstructing weep hole (W) entry.

⚠ CAUTION ⚠

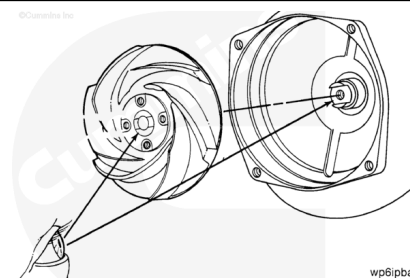
The seal will malfunction if the spring tension is not correct. One drop of Loctite® 290, or equivalent, must be applied on the joint between the seat and the shaft. More than one drop of the sealant can contact the sealing surfaces and cause the seal to malfunction.



Apply one drop **only** of Loctite® 290, or equivalent, to the seal and the parting line of the shaft.

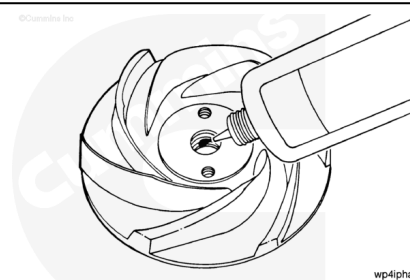
⚠ CAUTION ⚠

If the impeller and shaft have a slot for the drive key, the slots must be aligned before the impeller is pushed on the shaft.



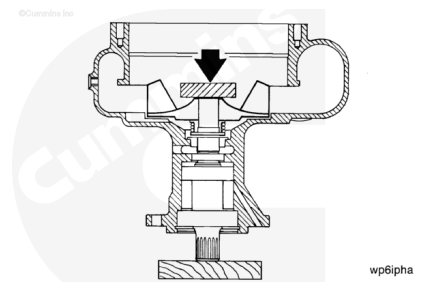
If either the impeller or shaft does **not** have a slot, omit this step. Position the impeller on the shaft so that the slots are aligned.

Apply a smooth coating of Loctite® 609, or equivalent, to the impeller bore.



Use an arbor press and a flat steel plate to support the shaft as shown.

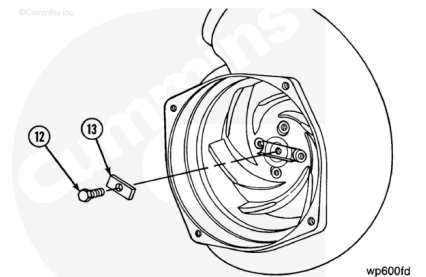
Install the impeller until it is even with the end of the shaft.



Note : If either the shaft or the impeller does not have a slot for the drive key, omit this step.

Install the drive key (13) and the self-locking capscrew (12).

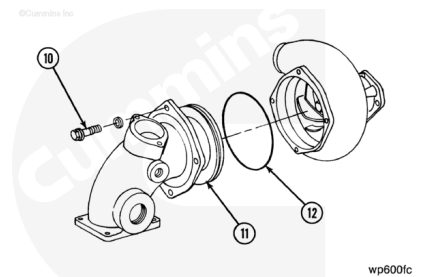
Torque Value: 20 n•m [177 in-lb]



Use vegetable oil to lubricate the o-ring.

Install the parts. Install the top and right capscrews.

Torque Value: 47 n•m [35 ft-lb]



Note : Do not tighten the capscrews for the bracket until the pump is installed on the engine.

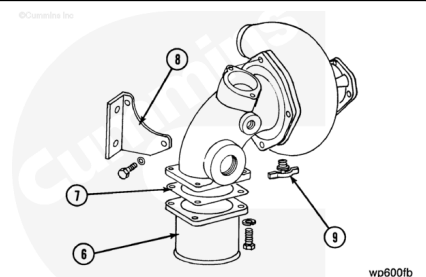
Install the support bracket (8) and capscrews.

Install the new gasket (7), connection (6), and capscrews.

Torque Value: 41 n•m [30 ft-lb]

Install the draincock (9).

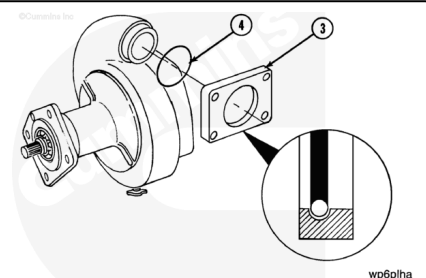
Torque Value: 15 n•m [133 in-lb]



Use vegetable oil to lubricate the new o-ring (4).

Install the o-ring in the groove of the adapter plate (3).

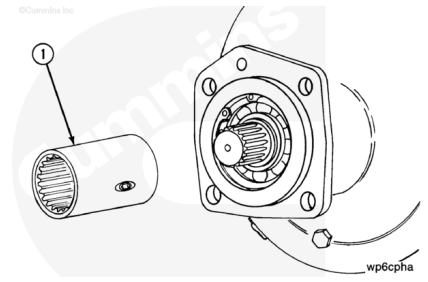
Install the adapter plate on the body. The thinner side of the plate **must** be nearer to the water pump body.



Use clean engine oil to lubricate the bearings while rotating the shaft.

Use clean engine oil to lubricate the splines on the shaft.

Install the spline coupling (1).



Dual Impeller

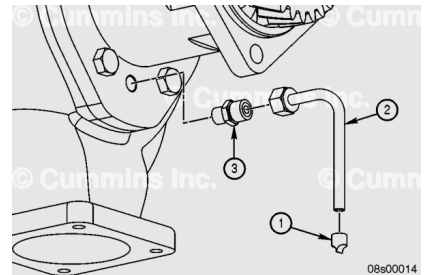
Install the weep hole drain tube fitting (3) onto the water pump and tighten.

Torque Value: 12 n•m [106 in-lb]

Install the weep hole drain tube (2) onto the fitting (3) and tighten the compression nut.

Torque Value: 12 n•m [106 in-lb]

Install the filter (1) onto the weep hole drain tube (2).



Install

Jacket

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

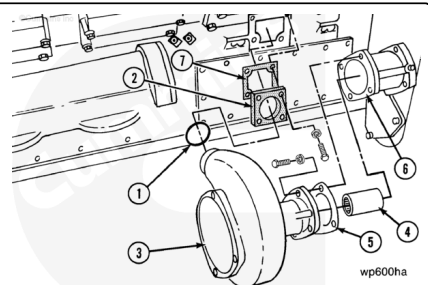
Install the o-ring (1) in the groove in the water pump adapter plate (2). Use vegetable oil to lubricate the o-ring.

Note : The adapter plate must be installed so the groove in the plate is toward the pump.

Install the adapter plate on the water pump (3).

Install the spline coupling (4) and the water pump shaft. Install the gasket (5) on the pilot on the water pump. Align the holes.

Position the pump on the water pump drive (6). Rotate the pump to align the spline coupling with the splines on the water pump drive shaft. Install the four lock washers and capscrews



that hold the drive to the pump. Tighten the capscrews **only** enough to hold the parts together.

Install the gasket (7) between the adapter plate and the cylinder block. Install, but do **not** tighten, the four lock washers and capscrews that hold the plate to the cylinder block.

Rotate the pump housing so that the pump outlet and the adapter plate are as close to the cylinder block as possible.

Tighten the four pump-to-pump drive capscrews.

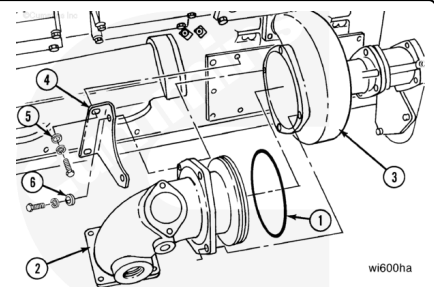
Torque Value: 60 n•m [44 ft-lb]

Tighten the four cylinder block capscrews to the adapter plate.

Torque Value: 40 n•m [30 ft-lb]

Install the o-ring (1) in the groove on the inlet housing (2). Use vegetable oil to lubricate the seal.

Install the housing in the bore in the water pump (3). Align the holes. Install the two lock washers and capscrews in the top and outside capscrew holes. Tighten the capscrews **only** enough to hold the parts together.



Install the bracket (4) to the pump using two large flat washers (6), lock washers and capscrews. Tighten the capscrews **only** enough to hold the bracket to the housing.

Install the two large flat washers (5), lock washers, and capscrews that hold the bracket to the hand hole cover. Tighten the capscrews **only** enough to hold the bracket to the hand hole cover.

Note : Make sure the bracket remains flat against both the housing and the hand cover while tightening the capscrews. If the bracket does not remain flat, it will malfunction and cause the pump or pump drive to also malfunction.

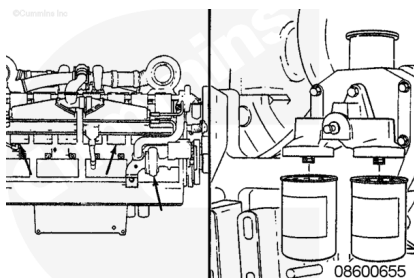
Tighten the four capscrews that hold the bracket to the inlet housing, and the inlet housing to the pump.

Torque Value: 40 n•m [30 ft-lb]

Tighten the capscrews that hold the bracket to the hand hole cover.

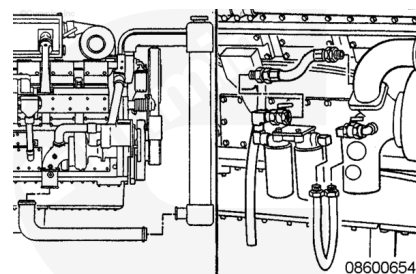
Torque Value: 60 n•m [44 ft-lb]

Replace the coolant filter. Refer to Procedure 008-006 in Section 8. (</qs3/pubsys2/xml/en/procedures/102/102-008-006-tr.html>)



Connect the lower radiator hose to the water pump inlet.

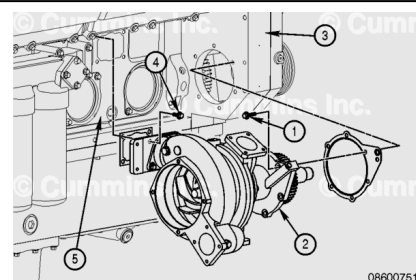
Connect the water filter lines.



Dual Impeller

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the water pump (2) into the gear housing (3). Use the six capscrews (1) and a new gasket.

Install the four capscrews (4) to the engine block (5). Use a new gasket.

Tighten the capscrews.

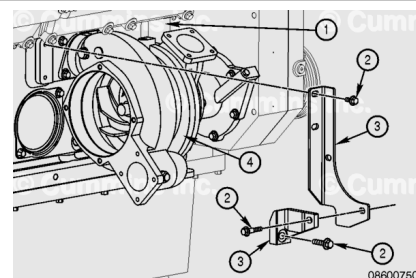
Torque Value:

Water Pump Mounting Capscrews 45 n•m [33 ft-lb]

Install the capscrews (2) and the water pump support bracket (3) to the water pump (4) and the engine block (1).

Install the two capscrews (2) to the water pump (4) and the water pump support bracket (3).

Install the two capscrews (2) to the engine block (1) and the water pump support bracket (3).



Torque Value:

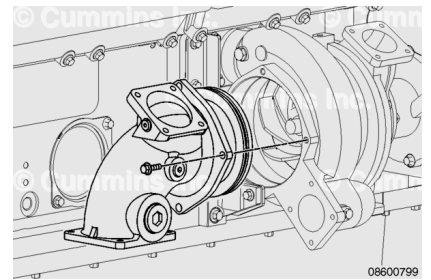
Support Bracket Capscrews 45 n•m [33 ft-lb]

Lubricate the new o-ring on the jacket water inlet.

Install the jacket water inlet and capscrews onto the water pump.

Torque Value:

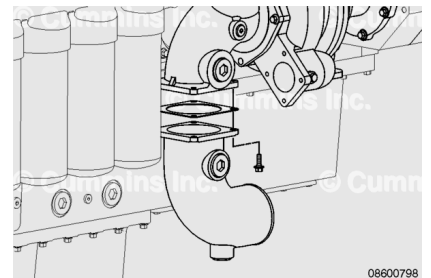
Jacket Water Inlet Capscrews 45 n•m [33 ft-lb]



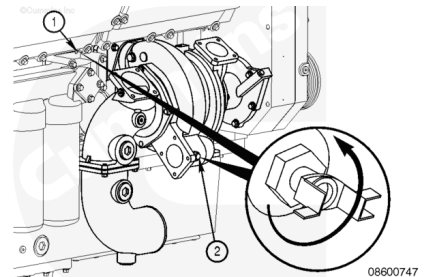
Install the jacket water radiator inlet connection, gasket, and four capscrews.

Torque Value:

Jacket Water Radiator Inlet Capscrews 45 n•m [33 ft-lb]



Close the draincock on the engine block (1) and on the water pump (2).



Finishing Steps

Jacket

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Install the coolant bypass tube. Refer to Procedure 008-005 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-005-tr.html)

- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Connect the battery cables. See equipment manufacturer service information.
- Operate the engine to 70°C [160°F] coolant temperature and check for leaks.

Dual Impeller

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Install the water pump bypass and feed tubes. Refer to Procedure 008-138 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-138.html)
- Connect the battery cables. See equipment manufacturer service information.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Install the alternator drive pulley. Refer to Procedure 009-010 in Section 9.
(/qs3/pubsys2/xml/en/procedures/28/28-009-010-tr.html)
- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13.
(/qs3/pubsys2/xml/en/procedures/28/28-013-005-tr.html)
- Operate the engine to 70°C [160°F] coolant temperature and check for leaks.